A Methodology for Building Human-Confabulated Hallucination Benchmarks

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Abstract

Every major hallucination benchmark generates its false content by prompting an LLM. Whether detection methods validated on such benchmarks still work when the false content is produced by a human has never been tested—because no methodology exists for constructing human-confabulated benchmarks at scale. We propose one. Our protocol operationalizes confabulation in the neuropsychological sense: a human non-expert writes plausible-sounding responses from memory, without consulting any source, while a verified grounded response serves as the paired control. The methodology is domain-agnostic, requires no specialized tooling, and produces pairs whose distributional properties differ fundamentally from LLM-generated benchmarks. We validate the protocol with a 212-pair dataset across nine domains and a three-way comparison showing that detection performance on an established benchmark (88–97%) does not transfer to human confabulations (69–78%). The distributional hypothesis explains why: sentence embeddings encode co-occurrence, not referential truth, so confabulations that stay within the register of their domain are invisible to cosine-similarity methods. We release the methodology, dataset, and companion tools to enable the community to build human-confabulated benchmarks at any scale.

1 Introduction

Article 15 of the EU AI Act [European Parliament, 2024] requires providers of high-risk AI systems to implement accuracy metrics. For retrieval-augmented generation (RAG) pipelines in healthcare, legal, and financial services, this means detecting when output is wrong. We use the term *confabulation*: the production of false information without intent to deceive [Berlyne, 1972], adopted for LLM outputs by Farquhar et al. [2024].

Detecting confabulation is an active area of research [Ji et al., 2023, Huang et al., 2025], and benchmarks are essential for measuring progress. The largest hallucination benchmarks—HaluEval [Li et al., 2023], HaluBench [Ravi et al., 2024], HalluLens [Bang et al., 2025]—generate false content by prompting LLMs. This raises a question that has not been empirically tested: does detection performance measured on LLM-generated false content transfer to false content produced by humans?

The question matters because the two processes are different. An LLM prompted to "generate a hallucinated answer" produces output conditioned on those instruction tokens, which enter the attention computation and influence every subsequent token [Vaswani et al., 2017]; a human non-expert confabulating from memory draws on incomplete knowledge and fills gaps with plausible-sounding material. Gudibande et al. [2023] showed that distributional properties of instruction-conditioned text are separable from factual content. The resulting text may differ in ways that affect detection—but without a human-confabulated benchmark, there is no way to check.

No such benchmark exists, and more importantly, no reproducible methodology for constructing one has been described. This paper provides both: a step-by-step protocol for producing human-confabulated question–response pairs, grounded in the neuropsychological literature

on confabulation [Berlyne, 1972, Moscovitch, 1995, Dalla Barba, 1993], and a 212-pair validation dataset demonstrating that the methodology produces confabulations with fundamentally different distributional properties from LLM-generated benchmarks.

2 Hallucination Benchmarks

The largest hallucination evaluation resources share a common construction method: an LLM generates the false content. HaluEval [Li et al., 2023] uses ChatGPT to produce 35,000 samples. HaluBench [Ravi et al., 2024] combines datasets with synthetic perturbations for 15,000 RAG triplets. HalluLens [Bang et al., 2025] generates dynamic tests from Wikipedia.

Alternative approaches avoid LLM-generated false content. FactScore [Min et al., 2023], RAGTruth [Niu et al., 2024], and FaithBench [Bao et al., 2025] evaluate natural model outputs, but cannot control for specific confabulation types and do not provide paired correct/false examples. TruthfulQA [Lin et al., 2022] targets widely shared misconceptions, not domain-specific confabulations. Sampling-based methods [Manakul et al., 2023, Kuhn et al., 2023, Farquhar et al., 2024] bypass benchmarks but require 5–20 generations per query.

What is missing is not merely a dataset but a *methodology*: a reproducible protocol for producing human-confabulated benchmarks in any domain, at any scale, that other research groups can adopt.

3 Confabulation Methodology

The methodology produces triplets (q, g, c): a domain-specific question, a verified grounded response, and a human confabulation. It operationalizes provoked confabulation in the neuropsychological sense [Berlyne, 1972, Moscovitch, 1995]: a question elicits a response from a person who does not know the answer but produces one anyway.

The construction pipeline is simple: select a domain and question, generate and verify a grounded response, ask a human to confabulate from memory without consulting any source, and check response lengths ($|g| \approx |c|$, $\pm 20\%$).

What makes the methodology non-trivial is not the pipeline but the confabulation mechanisms it produces. When a human confabulates from memory across different knowledge types [Dalla Barba, 1993] is using three main different mechanisms, each with different consequences for detection (Table 1).

3.1 Register preservation (regulatory/procedural domains)

Question: What is a Roth IRA?

Grounded: A Roth IRA is a US individual retirement account funded with after-tax dollars. Contributions are not tax-deductible, but qualified withdrawals in retirement are entirely tax-free. The annual contribution limit is \$7,000 (\$8,000 for age 50+), subject to income phase-outs.

Confabulated: A Roth IRA is a US employer-sponsored retirement vehicle that allows pre-tax payroll deductions up to 12% of gross income. The employer matches the first 4% dollar-for-dollar. Withdrawals before age 62 incur a 15% early-distribution surcharge, and all distributions are taxed as ordinary income.

In this example the confabulation inverts the tax treatment (after-tax $\rightarrow$ pre-tax), invents a matching rule, and fabricates every number—but stays entirely within the financial-regulatory register: "contributions," "distributions," "deductions," "surcharge" all co-occur naturally in financial text. Under the distributional hypothesis [Harris, 1954], the two responses are neighbors in embedding space as they share the same co-occurrence patterns. The difference is referential, not distributional [Bender and Koller, 2020].

Mechanism	Domain type	What the confabulator does	Distributional consequence
Register preservation	Regulatory, procedural	Replaces every factual claim—names, numbers, rules—while keeping domain-appropriate vocabulary and sentence structure intact.	$\cos(g, c)$ high. Confabulation is distributionally similar to grounded response.
Template-filling	Technical specification, coding	Preserves the syntactic template (name, classification, trigger, effect) while swapping content in the template slots.	$\cos(g, c)$ very high. Template dominates embedding; slot content contributes little.
Mechanism inversion	Declarative fact, scientific	Inverts or replaces the causal mechanism; each local transition is plausible but the global structure is wrong.	$\cos(g, c)$ moderate. Accumulated distributional deviation across the response produces a detectable shift.

Table 1: Confabulation mechanisms produced by the methodology. These are not design choices but emergent cognitive patterns observed when humans confabulate from memory across different knowledge types. The distributional consequence column describes the prediction under the distributional hypothesis [Harris, 1954]: detection by embedding similarity depends on whether the mechanism moves the confabulation outside the distributional register of its domain.

3.2 Template-filling (technical specification domains)

Question: What does `GROUP BY` do in SQL?

Grounded: GROUP BY is a SQL clause that arranges identical data into groups. It is used with aggregate functions (COUNT, SUM, AVG, MAX, MIN) to perform calculations on each group. Rows sharing the same values in specified columns are collapsed into a single summary row.

Confabulated: GROUP BY is a SQL clause that partitions query execution across multiple threads. When applied to a column, it instructs the database engine to split the table into segments based on cardinality and process each in parallel. It is paired with MERGE functions to recombine the parallel results.

Both responses follow the same template: clause name, what it does, how it works, what it pairs with. Both use SQL register vocabulary ("clause," "column," "table," "query"). The template determines the embedding; the content filling the slots—where truth resides—contributes little. This is why template-structured domains are the hardest for cosine-similarity detection: the confabulation is structurally identical to the grounded response.

3.3 Mechanism inversion (declarative-knowledge domains)

Question: What is CRISPR-Cas9?

Grounded: CRISPR-Cas9 is a gene-editing tool adapted from a bacterial immune system. Bacteria capture snippets of viral DNA and store them as "spacers" between repeated sequences. The Cas9 protein, guided by a synthetic RNA, cuts both strands of DNA at a precise location, allowing researchers to delete, replace, or insert genetic material.

Confabulated: CRISPR-Cas9 is a protein-folding correction mechanism originally discovered in archaea. The Cas9 enzyme identifies misfolded protein sequences by binding to their mRNA transcripts before translation. It recruits chaperone molecules to refold the protein correctly. Researchers have adapted this system to fix genetic mutations by targeting the mRNA stage rather than the DNA itself.

This confabulation stays within the molecular biology register ("protein," "mRNA," "tran-

scripts," "chaperone") but invents an entirely different mechanism (protein-folding correction vs. gene editing). Each local transition is distributionally plausible, but the accumulated vocabulary shift across the full response—from gene-editing terms to protein-folding terms—moves the embedding away from the grounded response. This mechanism is the most detectable of the three. The confabulator does not know the real mechanism, so they invent one from adjacent knowledge—protein folding instead of gene editing. The invented mechanism brings its own vocabulary, and that vocabulary occupies a different region of embedding space than the grounded response. The further the substituted mechanism is from the real one, the larger the distributional shift and the easier the detection.

3.4 How human confabulation differs from LLM-generated hallucination

When an LLM is instructed to "generate a hallucinated response," the instruction tokens condition the entire output distribution [Vaswani et al., 2017]. Gudibande et al. [2023] showed that distributional properties of instruction-conditioned text are separable from factual content. A person confabulating follows a different cognitive process: she attempts to answer from memory, and where memory fails, she builds plausible-sounding material from partial knowledge and analogical reasoning.

The three mechanisms are not prescribed by the protocol. They emerge naturally when a person confabulates across different knowledge types — the protocol simply creates the conditions for them to appear. The validation experiments (Section 6) confirms that these mechanisms produce confabulations with different distributional properties than LLM prompting.

4 Why Distributional Similarity Cannot Detect In-Register Confabulation

Sentence transformers map text onto the unit hypersphere $\mathbb{S}^{d-1}$ through contrastive training [Reimers and Gurevych, 2019, Wang and Isola, 2020]. What determines proximity on this surface is co-occurrence: words that appear in similar contexts receive similar representations [Harris, 1954]. Truth plays no role. Two responses that use the same vocabulary in the same syntactic patterns will be neighbours in embedding space whether their factual claims are correct or not.

This creates a blind spot. A confabulation that preserves the vocabulary and syntax of its domain — staying in-register — will land near the grounded response. The difference between them is referential: one corresponds to something real, the other does not. But referential correspondence requires access to the world beyond the text [Bender and Koller, 2020], and embeddings encode only form.

How large this blind spot is depends on the domain. In template-structured domains (coding, SQL), the syntactic template determines the embedding; the content filling the slots — where truth resides — contributes little. In declarative-knowledge domains, a confabulator with incomplete knowledge may introduce vocabulary outside the register, producing a distributional anomaly that embeddings can detect. This is not a limitation of any particular model. It follows from what distributional representations encode: co-occurrence structure [Mardia and Jupp, 2000], not reference.

5 Validation Experiment

We validate the methodology by applying it to generate a 212-pair dataset across nine domains (Table 2), then characterizing the output through a three-way comparison.

Domain	n	Knowledge type
Python coding	47	Template-structured
Finance	40	Regulatory / procedural
Medical	40	Clinical / declarative
Science	21	Declarative fact
TypeScript coding	18	Template-structured
History	14	Declarative fact
Law	11	Regulatory / procedural
General knowledge	11	Mixed
Geography	10	Declarative fact
Total	**212**	

Table 2: Validation dataset: domain distribution. Response lengths are comparable (grounded: $\mu = 55.6$ words; confabulated: $\mu = 57.3$). Per-domain samples for $n < 15$ domains are reported descriptively.

5.1 Three-way comparison

We compare detection difficulty across three conditions:

1. **Human confabulations** ($n = 212$): produced by the methodology in Section 3.
2. **LLM confabulations** ($n = 212$): an LLM generates false responses to the same 212 questions, simulating the construction method of existing benchmarks. Questions and grounded responses are identical to Dataset 1.
3. **HaluEval QA** ($n = 500$): a random sample from HaluEval QA Li et al. [2023], an established LLM-generated hallucination benchmark.

5.2 Models and metrics

We test four sentence transformers: all-MiniLM-L6-v2 ($384d$), all-mpnet-base-v2 ($768d$), bge-small-en-v1.5 ($384d$), and gte-small ($384d$). All embeddings are ℓ_2-normalized.

We use three metrics for response comparison:

- Paired similarity, $\cos(g, c)$ for the same question.
- Detection accuracy, the fraction of pairs where $\cos(q, g) > \cos(q, c)$ (random baseline: 50%).
- Mean Δ, the average $\cos(q, g) - \cos(q, c)$. In prior work [Marín, 2025], we formalized a context-anchored version of this metric as the Semantic Grounding Index (SGI); the present experiment measures the context-free setting.

5.3 Statistical methods

We quantify uncertainty around each detection rate with Wilson score 95% confidence intervals [Wilson, 1927]. To test whether detection is significantly above random guessing, we use one-sided binomial exact tests (H_0: detection = 50%), applying Holm–Bonferroni correction when comparing across domains. Effect sizes are reported as Cohen's h. For direct human-vs.-LLM comparisons on the same questions we use Wilcoxon signed-rank tests, and for the template-vs.-declarative split we use Fisher's exact test.

Model	Response	Det%	95% CI	Δ	Paired
all-MiniLM-L6-v2	Human	69.3%	[62.8, 75.2]	+.044	.722
	LLM	72.6%	[66.3, 78.2]	+.027	.858
	HaluEval	95.4%***	[93.2, 96.9]	+.409	.114
all-mpnet-base-v2	Human	78.3%	[72.3, 83.3]	+.049	.766
	LLM	75.9%	[69.8, 81.2]	+.028	.888
	HaluEval	97.2%***	[95.4, 98.3]	+.414	.104
bge-small-en-v1.5	Human	71.7%	[65.3, 77.3]	+.024	.845
	LLM	73.1%	[66.8, 78.6]	+.013	.918
	HaluEval	96.6%***	[94.6, 97.9]	+.265	.453
gte-small	Human	69.3%	[62.8, 75.2]	+.011	.917
	LLM	73.6%	[67.3, 79.1]	+.006	.955
	HaluEval	88.4%***	[85.3, 90.9]	+.098	.777

Table 3: Three-way detection comparison. Det%: fraction where $\cos(q, g) > \cos(q, c)$. CI: Wilson 95%. Δ: mean $\cos(q, g) - \cos(q, c)$. Paired: mean $\cos(g, c)$. $^{***}p < .001$ vs. Human (Wilcoxon signed-rank). All human and LLM rates significantly above 50% ($p < 10^{-8}$, binomial).

6 Results

6.1 Three-way comparison

Table 3 reports the three-way comparison. Two patterns are consistent across all four models.

Detection does not transfer. On HaluEval, detection reaches 88–97%; on human confabulations, it drops to 69–78%. The reason is visible in paired similarity: HaluEval's correct and hallucinated answers differ sharply (cosine 0.10–0.78), while both confabulation sets remain close to their grounded counterparts (0.72–0.96). In other words, HaluEval hallucinations are easy to detect because they look different from the correct text, not because the method captures factual error.

Human and LLM confabulations are comparably difficult. Detection on human confabulations (69–78%) and LLM confabulations (73–76%) is similar. Yet the two sets differ underneath: LLM confabulations sit closer to the grounded responses (paired similarity 0.86–0.96 vs. 0.72–0.92, smaller Δ). That comparable detection rates arise from different distributional profiles reflects an inverted domain pattern, analyzed below.

6.2 Per-domain analysis

Tables 4 and 5 show the domain-level and aggregate results. Template-structured domains show detection at 56.9% (95% CI [44.8%, 68.2%]) for human confabulations, not significantly above random guessing ($p = .161$). Declarative-knowledge domains show 87.8% (95% CI [81.5%, 92.1%] with $p < 10^{-15}$). Fisher's exact test confirms the split ($p = 1.5 \times 10^{-6}$), consistent across all four models.

The asymmetry between human and LLM confabulations is informative. In declarative domains, human confabulations are easy to detect (87.8% vs. 81.0%) because the human non-expert fabricates off-distribution content (fictitious institutions, impossible mechanisms). In template domains, human confabulations are more difficult to detect, with the LLM introducing minor stylistic fabrications.

This pattern validates the methodology: confabulations produced by the protocol have different distributional properties from those produced by prompting an LLM, and these differences

Domain	n	Det%	95% CI	Adj. p	
Template-structured domains					
Python coding	47	55.3%	[41.2, 68.6]	.280	ns
TypeScript coding	18	61.1%	[38.6, 79.7]	.481	ns
Declarative-knowledge domains					
Finance	40	95.0%	[83.5, 98.6]	$< .001$	***
Science	21	95.2%	[77.3, 99.2]	$< .001$	***
History	14	100.0%	[78.5, 100.0]	$< .001$	***
Law	11	100.0%	[74.1, 100.0]	.003	**
Medical	40	75.0%	[59.8, 85.8]	.006	**
General	11	81.8%	[52.3, 94.9]	.131	ns
Geography	10	70.0%	[39.7, 89.2]	.516	ns

Table 4: Per-domain detection for human confabulations (all-mpnet-base-v2). CI: Wilson 95%. Adj. p: Holm-Bonferroni corrected. Domains with $n < 15$: CIs span 26–50 pp.

Category	Dataset	n	Det%	95% CI	p
Template	Human	65	56.9%	[44.8, 68.2]	.161
	LLM	65	64.6%	[52.5, 75.1]	.012
Declarative	Human	147	87.8%	[81.5, 92.1]	$< 10^{-15}$
	LLM	147	81.0%	[73.8, 86.5]	$< 10^{-14}$
Fisher's exact test (template vs. declarative):					
	Human				1.5×10^{-6}
	LLM				.014

Table 5: Aggregate detection by domain category (all-mpnet-base-v2). The template–declarative split is significant for both dataset types.

are domain-dependent in ways predicted by the distributional account of Section 4.

7 Discussion and Limitations

We have presented a methodology for constructing hallucination benchmarks from human confabulations rather than LLM-generated content. The 212-pair dataset validates the protocol; the contribution is the methodology itself, which can be applied to any domain at any scale.

The validation experiment shows that detection performance on LLM-generated benchmarks does not transfer to confabulations produced by this methodology. The gap in paired similarity between HaluEval and our confabulation sets is consistent with multiple explanations—instruction-conditioned generation, task design, prompt specificity—and distinguishing among them would require interpretability analysis beyond the scope of this work. The experiment measures cosine similarity without external context or model internals; methods using retrieved context [Marín, 2025], multiple samples [Manakul et al., 2023, Farquhar et al., 2024], or model internal states may perform differently.

Validation dataset comes from a single confabulator (person-specific patterns), English only, per-domain n too small for domain-level inference after correction (power analysis: $n \approx 286$ per domain for 80% power). These are limitations of the validation, not of the methodology—which is designed to be applied by multiple generators across languages and domains.

8 Conclusion

We present a reproducible methodology for generating human-confabulated hallucination benchmarks, validated with a 212-pair dataset. The protocol produces confabulations whose distributional properties differ from LLM-generated benchmarks: detection drops from 88–97% to 69–78%. The template–declarative split ($p = 1.5 \times 10^{-6}$) confirms that detectability depends on whether a confabulation stays within the register of its domain, as the distributional hypothesis predicts [Harris, 1954]. We release the methodology, dataset, and companion LLM-confabulated control.[1]

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